

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (Comm. System Engg.) Sem - II University Examination, 2010-11

EC – 710 Digital Image and Speech Processing

Date: 18.05.11, Wednesday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

- Instructions:**
- (1) All questions are compulsory.
 - (2) Indicate clearly, the option you attempt along with its respective question number.
 - (3) Figures to right indicate marks
 - (4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

- Q-1 A** A 4 X 4, 4bits/pixel image passed through a point wise intensity transformation given by $s = T(r) = \alpha \log_2(1+r) + \beta$. Where α and β are unknown parameters. Only few pixels are available in the input and output image as shown in figure Q1-A. [4]

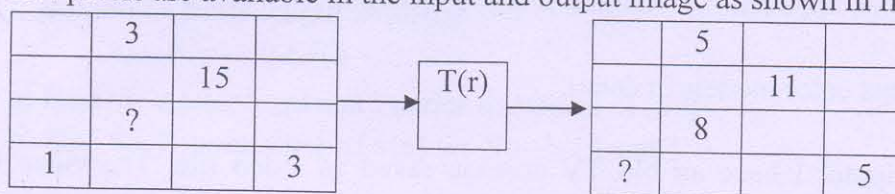


Figure Q1-A

- (a) Find α and β .
- (b) What is the value of pixel with the “?” mark in the **output** image.
- (c) What is the value of pixel with the “?” mark in the **input** image.

- B** Below is a histogram of 4-bit image. [7]

r	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
H(r)	0	0	50	100	150	100	0	0	0	0	0	50	50	0	0	0

Find the transformation function $s=T(r)$ that will equalize the histogram.

- Q-2 A** Consider the 3X3 Prewitt operator P, shown in figure Q2-A. Assume that it is applied to an image I composed of vertical step edge as shown in figure Q2-B. [6]

-1	0	1
-1	0	1
-1	0	1

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0

Figure Q2-A

Figure Q2-B

- 1) Compute the result of correlating P with I.
- 2) Compute the result of convolving P with I.

- 3) Let A be the 3X3 smoothing operator composed of all 1's. If we first smooth the image by correlating it with A and then correlate with P, is that the same as correlating with P then with A? Why or why not?

OR

- A Describe the Adaptive median filtering in detail. [6]
- B Explain the concept of Adjacency, Connectivity, Region and Boundaries. Give suitable example. [4]

OR

- B Suppose that you form a low pass spatial filter that averages the four immediate neighbors of point (x,y) but excludes the point itself. [4]
- 1) Find the equivalent filter $H(u,v)$ in the frequency domain.
 - 2) Show that your result is low pass filter.

- C List the various application of Digital Image Processing. [3]

- Q-3 A Define various color models in detail. [7]

- B On my computer I have an old TV episode saved as video file. The video is composed of images that are 512 X384 pixels, where each pixel is 3 bytes or 24 bits. The episode is 50 minutes, 40 second long. Images play at a rate of 23 frames / second. With no compression, how big this file would be? [4]

OR

- Q-3 A Determine the DFT of the following image. [7]

0	1	2	1
1	2	3	2
2	3	4	3
1	2	3	2

- B What is Pseudo color Image processing? Explain one of the techniques of Pseudo color image processing. [4]

SECTION – II

Q-4 A Define: Phonemes, Pitch and Speech signal [3]

B Describe: “Contribution of DSP in Speech processing applications”. [5]

C Consider the sequence [3]

$$x(n) = a^n \quad n \geq n_0 \\ = 0 \quad n < n_0$$

- 1) Find the z-transform of $x(n)$.
- 2) Find the Fourier transform of $x(n)$. Under what conditions does the Fourier transform exist?

Q-5 A Explain: Flanagan’s speech production mechanism. [4]

B Consider the finite length sequence [4]

$$x(n) = \delta(n) + 0.5\delta(n-5)$$

- a) Find the z-transform and Fourier transform of $x(n)$.
- b) Find the N-point DFT of $x(n)$ for $N=10$ and 5 .

C Explain: General discrete-time model for speech production. Draw the neat sketch. [4]

OR

Q-5 A Derive the complete flow diagram of a two tube model. [4]

B A commonly used approximation to the glottal pulse is [8]

$$g(n) = na^n \quad n \geq 0 \\ = 0 \quad n < 0$$

- 1) Sketch the Fourier transform, $G(e^{j\omega})$, as a function of ω .
- 2) Show how a should be chosen that

$$20 \log_{10} |G(e^{j0})| - 20 \log_{10} |G(e^{j\pi})| = 60 \text{ dB}.$$

Q-6 A Describe: “Pitch period estimation using a parallel processing approach”. [6]

B Write on “Principles of Linear Predictive Analysis”. [6]

OR

Q-6 A Explain: Delta modulator Codec with adaptive step size. [6]

B Describe procedure of Filter bank design using IIR filters. [6]